

Eight edge cases on the A10 deterministic board

7 by 5 board; goal rows 1, 2, 3; simultaneous U/D/L/R; deterministic carrier-wins-contests resolution. Players never occupy the same square. The loser of a contested square or swap receives the ball.

Selection rule: preserve the best worst-case payoff, then maximize mean payoff over opponent actions among exact security ties. Identical-score ties use action order. The secondary criterion is a documented selection convention, not a claim that Nash equilibrium is unique. No uniform tie mix is substituted for the solver policy.

Case 1 selects Down for player 0: it weakly dominates Up and Left. The four zeros against Right describe optimal continuation from four new states, not a defender committed to Right forever. These are discounted stationary values with gamma 0.9, not the undiscounted 100-turn assignment.

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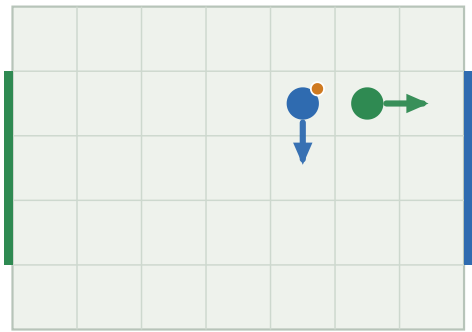
Case 1: The swap trap

State (4, 3, 5, 3, 0); player 0 carries the ball. V = +0.000000.

Selected policy: P0 D 100%; P1 R 100%. Security-tied alternatives: P0 U/D/L; P1 R. Alternatives are not probabilities.

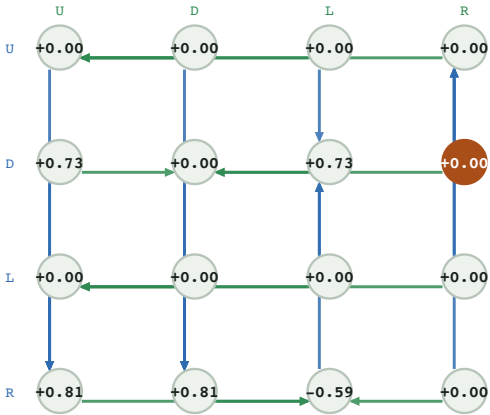
player 0's U/D/L never touch player 1 at all this turn, so all three guarantee at least 0. D weakly dominates U and L: it earns 0.729 against U/L, while U and L earn zero against every reply. The refined policy is D, not a uniform tie mix. Only R reaches player 1's square, and against L it triggers a genuine swap (verified below: (R, L) -> (5,3,4,3,1)) that hands player 1 the ball, dragging R's own worst case to -0.59. Player 1's unique safe column is R for an unrelated reason: U/D/L each let a smart player 0 reach a good, independent square elsewhere in the matrix (0.81/0.81/0.729) that has nothing to do with the swap cell.

The swap trap



pure equilibrium · V = +0.000

Q matrix -- (4, 3, 5, 3, 0)



pure saddle at D/R

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.729000	0.000000	0.729000	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.810000	0.810000	-0.590490	0.000000

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(4, 4, 5, 4, 0)	(4, 4, 5, 2, 0)	(4, 4, 4, 3, 0)	(4, 4, 6, 3, 0)
D	(4, 2, 5, 4, 0)	(4, 2, 5, 2, 0)	(4, 2, 4, 3, 0)	(4, 2, 6, 3, 0)
L	(3, 3, 5, 4, 0)	(3, 3, 5, 2, 0)	(3, 3, 4, 3, 0)	(3, 3, 6, 3, 0)
R	(5, 3, 5, 4, 0)	(5, 3, 5, 2, 0)	(5, 3, 4, 3, 1)	(5, 3, 6, 3, 0)

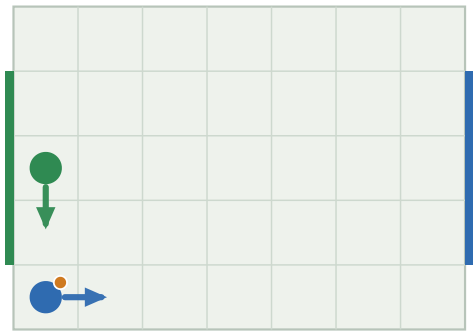
Case 2: Pinned in the corner

State (0, 0, 0, 2, 0); player 0 carries the ball. V = +0.000000.

Selected policy: P0 R 100%; P1 D 100%. Security-tied alternatives: P0 D/L/R; P1 D/R. Alternatives are not probabilities.

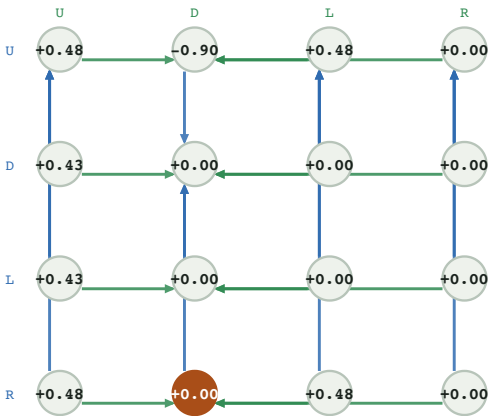
player 0's U is the one risky action: it targets the empty cell (0,1), which player 1's D also targets. Neither player is standing there, so it's an open (non-blocked) contest -- player 0 physically wins the race ((U, D) -> (0,1,0,2,1)) but the A10 rule still gives the ball to the loser regardless of who reached the cell first, crashing the value to -0.9. D, L, and R never reach that cell, so they tie safely at 0.

Pinned in the corner



pure equilibrium · V = +0.000

Q matrix -- (0, 0, 0, 2, 0)



pure saddle at R/D

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.478297	-0.900000	0.478297	0.000000
D	0.430467	0.000000	0.000000	0.000000
L	0.430467	0.000000	0.000000	0.000000
R	0.478297	0.000000	0.478297	0.000000

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(0, 1, 0, 3, 0)	(0, 1, 0, 2, 1)	(0, 1, 0, 2, 0)	(0, 1, 1, 2, 0)
D	(0, 0, 0, 3, 0)	(0, 0, 0, 1, 0)	(0, 0, 0, 2, 0)	(0, 0, 1, 2, 0)
L	(0, 0, 0, 3, 0)	(0, 0, 0, 1, 0)	(0, 0, 0, 2, 0)	(0, 0, 1, 2, 0)
R	(1, 0, 0, 3, 0)	(1, 0, 0, 1, 0)	(1, 0, 0, 2, 0)	(1, 0, 1, 2, 0)

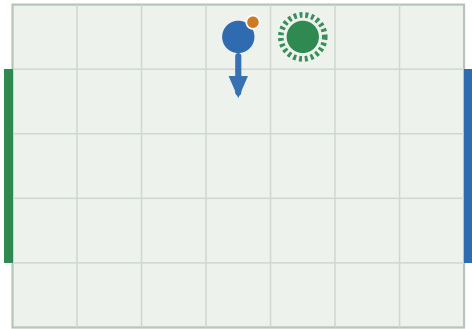
Case 3: The standoff

State (3, 4, 4, 4, 0); player 0 carries the ball. V = +0.000000.

Selected policy: P0 D 100%; P1 U 100%. Security-tied alternatives: P0 U/D/L; P1 U/D/R. Alternatives are not probabilities.

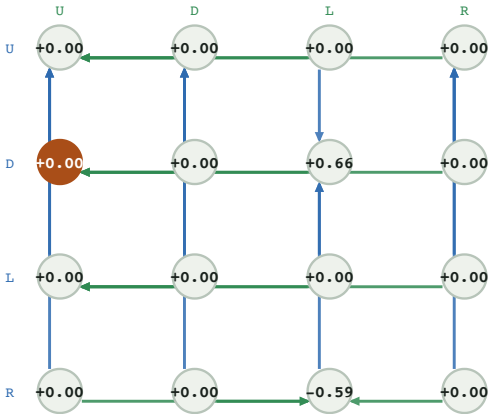
both players are already on the top row, one cell apart. U is wall-clamped (a genuine no-op) for both -- not a real move upward. R is player 0's one risky action (it reaches player 1's square: blocked against U, a real swap against L); L is player 1's mirror risk. D never touches the opponent for either player, so U/D/L (player 0) and U/D/R (player 1) all tie at the exact same safe value.

The standoff



pure equilibrium · V = +0.000

Q matrix -- (3, 4, 4, 4, 0)



pure saddle at D/U

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.000000	0.000000	0.656100	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.000000	0.000000	-0.590490	0.000000

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(3, 4, 4, 4, 0)	(3, 4, 4, 3, 0)	(3, 4, 4, 4, 1)	(3, 4, 5, 4, 0)
D	(3, 3, 4, 4, 0)	(3, 3, 4, 3, 0)	(3, 3, 3, 4, 0)	(3, 3, 5, 4, 0)
L	(2, 4, 4, 4, 0)	(2, 4, 4, 3, 0)	(2, 4, 3, 4, 0)	(2, 4, 5, 4, 0)
R	(3, 4, 4, 4, 1)	(4, 4, 4, 3, 0)	(4, 4, 3, 4, 1)	(4, 4, 5, 4, 0)

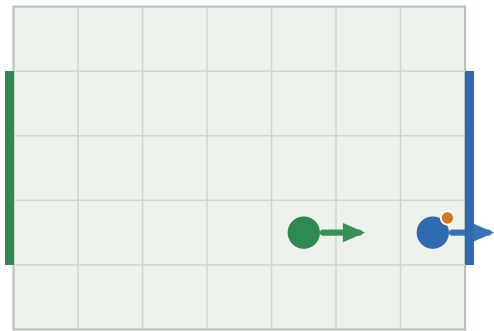
Case 4: The open goal

State (6, 1, 4, 1, 0); player 0 carries the ball. V = +1.000000.

Selected policy: P0 R 100%; P1 R 100%. Security-tied alternatives: P0 R; P1 U/D/L/R. Alternatives are not probabilities.

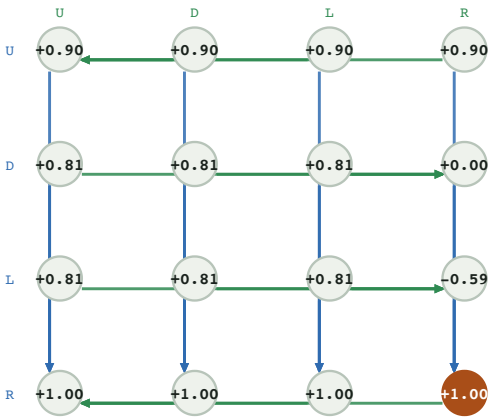
player 0 already sits at the right wall in a goal row, with the ball. Off a goal row this exact same boundary push would just hold in place -- here it scores instead: verified, game.transitions() returns a terminal player-0 win for R against every one of player 1's replies. R is the unique optimum (maximin 1.0); player 1's columns all tie at minimax 1.0 too, because there is nothing any of them can do about it.

The open goal



pure equilibrium · V = +1.000

Q matrix -- (6, 1, 4, 1, 0)



pure saddle at R/R

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.900000	0.900000	0.900000	0.900000
D	0.810000	0.810000	0.810000	0.000000
L	0.810000	0.810000	0.810000	-0.590490
R	1.000000	1.000000	1.000000	1.000000

All 16 successor states, from the transition engine

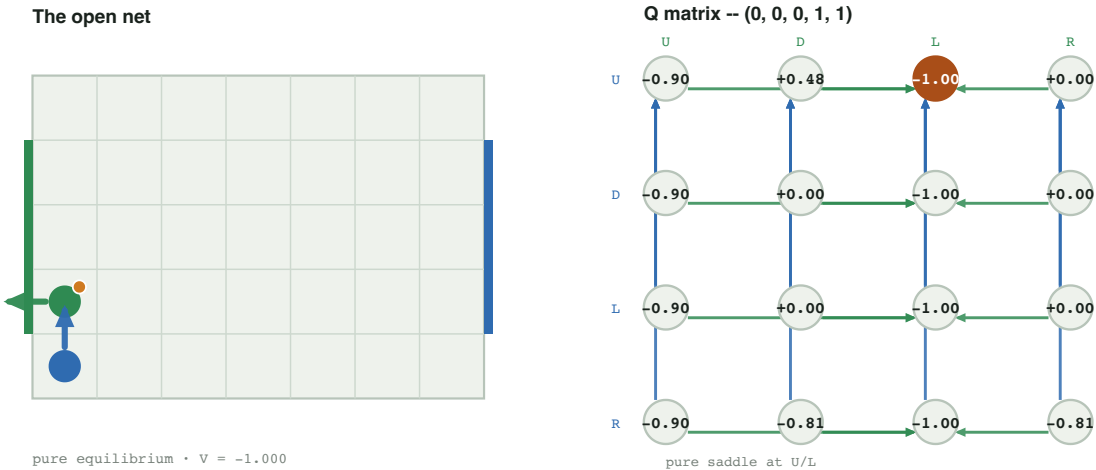
P0 / P1	U	D	L	R
U	(6, 2, 4, 2, 0)	(6, 2, 4, 0, 0)	(6, 2, 3, 1, 0)	(6, 2, 5, 1, 0)
D	(6, 0, 4, 2, 0)	(6, 0, 4, 0, 0)	(6, 0, 3, 1, 0)	(6, 0, 5, 1, 0)
L	(5, 1, 4, 2, 0)	(5, 1, 4, 0, 0)	(5, 1, 3, 1, 0)	(5, 1, 4, 1, 1)
R	(-1, -1, -1, -1, 0)	(-1, -1, -1, -1, 0)	(-1, -1, -1, -1, 0)	(-1, -1, -1, -1, 0)

Case 5: The open net

State (0, 0, 0, 1, 1); player 1 carries the ball. $V = -1.000000$.

Selected policy: P0 U 100%; P1 L 100%. Security-tied alternatives: P0 U/D/L/R; P1 L. Alternatives are not probabilities.

the mirror of the open goal from player 1's side: player 1 is at the left wall in a goal row, ball in hand, and L scores unconditionally (verified against every reply). The bleaker half of this case is player 0's row: every one of U/D/L/R ties at the exact same -1.0 -- there is nothing player 0 can do differently that changes the outcome, which is itself worth seeing on a board rather than only in a number.



Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	-0.900000	0.478297	-1.000000	0.000000
D	-0.900000	0.000000	-1.000000	0.000000
L	-0.900000	0.000000	-1.000000	0.000000
R	-0.900000	-0.810000	-1.000000	-0.810000

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(0, 1, 0, 2, 1)	(0, 1, 0, 0, 0)	(-1, -1, -1, -1, 1)	(0, 1, 1, 1, 1)
D	(0, 0, 0, 2, 1)	(0, 0, 0, 1, 0)	(-1, -1, -1, -1, 1)	(0, 0, 1, 1, 1)
L	(0, 0, 0, 2, 1)	(0, 0, 0, 1, 0)	(-1, -1, -1, -1, 1)	(0, 0, 1, 1, 1)
R	(1, 0, 0, 2, 1)	(1, 0, 0, 0, 1)	(-1, -1, -1, -1, 1)	(1, 0, 1, 1, 1)

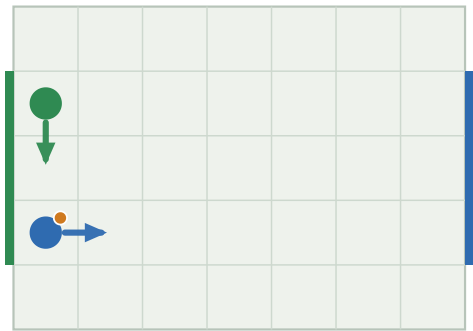
Case 6: The getaway

State (0, 1, 0, 3, 0); player 0 carries the ball. V = +0.531441.

Selected policy: P0 R 100%; P1 D 100%. Security-tied alternatives: P0 R; P1 U/D/L/R. Alternatives are not probabilities.

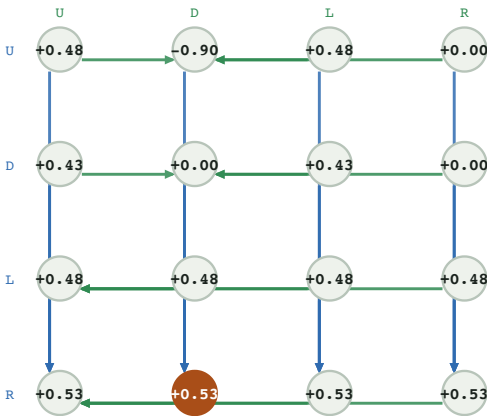
R is player 0's *unique* safe action (0.531441, no tie) -- an entirely independent move to (1,1) that never depends on player 1's reply. U looks just as reasonable but is a trap: against D, both target the same empty cell (0,2); player 0 wins the race there (verified: (U, D) -> (0,2,0,3,1)) and immediately loses the ball anyway, crashing to -0.9. Player 1's columns all tie at 0.531441 -- there is no reply that slows the escape down.

The getaway



pure equilibrium · V = +0.531

Q matrix -- (0, 1, 0, 3, 0)



pure saddle at R/D

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.478297	-0.900000	0.478297	0.000000
D	0.430467	0.000000	0.430467	0.000000
L	0.478297	0.478297	0.478297	0.478297
R	0.531441	0.531441	0.531441	0.531441

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(0, 2, 0, 4, 0)	(0, 2, 0, 3, 1)	(0, 2, 0, 3, 0)	(0, 2, 1, 3, 0)
D	(0, 0, 0, 4, 0)	(0, 0, 0, 2, 0)	(0, 0, 0, 3, 0)	(0, 0, 1, 3, 0)
L	(0, 1, 0, 4, 0)	(0, 1, 0, 2, 0)	(0, 1, 0, 3, 0)	(0, 1, 1, 3, 0)
R	(1, 1, 0, 4, 0)	(1, 1, 0, 2, 0)	(1, 1, 0, 3, 0)	(1, 1, 1, 3, 0)

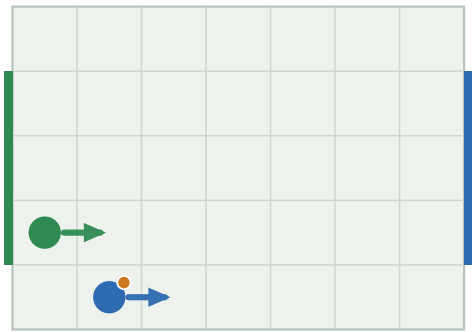
Case 7: The tightest tie

State (1, 0, 0, 1, 0); player 0 carries the ball. V = +0.000000.

Selected policy: P0 R 100%; P1 R 100%. Security-tied alternatives: P0 D/R; P1 R. Alternatives are not probabilities.

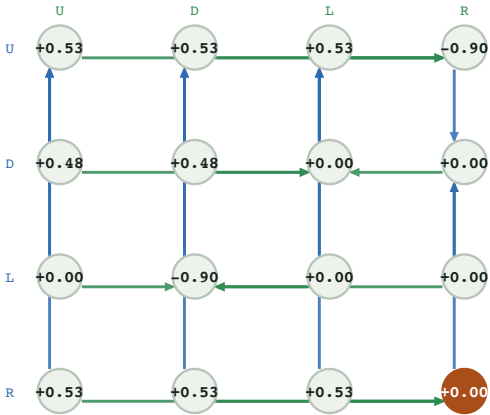
no state on this entire board ever has a fully unique best action for *both* players simultaneously -- checked exhaustively, (1,1) never occurs among all 2380 states. This is the smallest tie shape that does occur (only 16 of 2380 states): player 0's D and R tie exactly; player 1's R is uniquely optimal. U and L each look just as safe as D/R, but each hides its own contest: (U, R) both target the same cell (1,1), and (L, D) both target (0,0) -- verified below, the carrier wins the square but the ball still goes to the defender either way -- while D and R never target the same cell as player 1 at all.

The tightest tie



pure equilibrium · V = +0.000

Q matrix -- (1, 0, 0, 1, 0)



pure saddle at R/R

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.531441	0.531441	0.531441	-0.900000
D	0.478297	0.478297	0.000000	0.000000
L	0.000000	-0.900000	0.000000	0.000000
R	0.531441	0.531441	0.531441	0.000000

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(1, 1, 0, 2, 0)	(1, 1, 0, 0, 0)	(1, 1, 0, 1, 0)	(1, 1, 0, 1, 1)
D	(1, 0, 0, 2, 0)	(1, 0, 0, 0, 0)	(1, 0, 0, 1, 0)	(1, 0, 1, 1, 0)
L	(0, 0, 0, 2, 0)	(0, 0, 0, 1, 1)	(0, 0, 0, 1, 0)	(0, 0, 1, 1, 0)
R	(2, 0, 0, 2, 0)	(2, 0, 0, 0, 0)	(2, 0, 0, 1, 0)	(2, 0, 1, 1, 0)

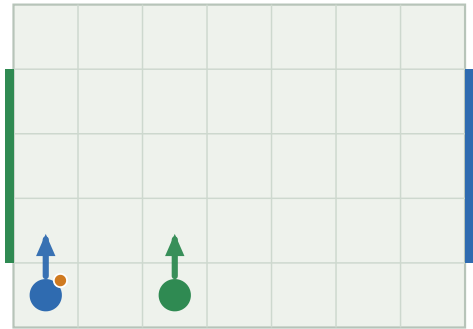
Case 8: The dead zone

State (0, 0, 2, 0, 0); player 0 carries the ball. V = +0.000000.

Selected policy: P0 U 100%; P1 U 100%. Security-tied alternatives: P0 U/D/L/R; P1 U/D/L/R. Alternatives are not probabilities.

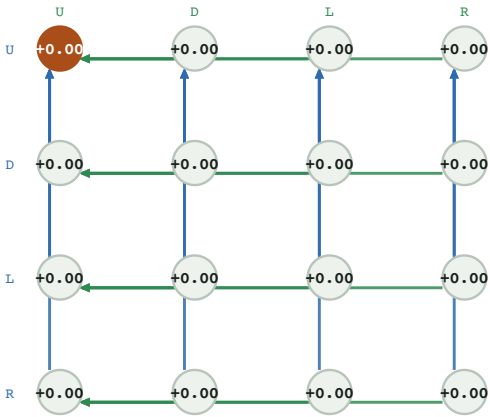
the single most common shape on the whole board (514 of 2380 states): both players are fully indifferent among all four actions. It is not merely that the guaranteed values tie -- every one of the 16 cells in the Q matrix is exactly 0.0, including a genuine ball-swapping contest ((R, L) -> a state that is itself also worth exactly 0, verified below). All successors have the same optimal-continuation value, although their physical positions differ.

The dead zone



pure equilibrium · V = +0.000

Q matrix -- (0, 0, 2, 0, 0)



pure saddle at U/U

Q: player 0 payoff, optimal continuation after this turn

P0 / P1	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.000000	0.000000	0.000000	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.000000	0.000000	0.000000	0.000000

All 16 successor states, from the transition engine

P0 / P1	U	D	L	R
U	(0, 1, 2, 1, 0)	(0, 1, 2, 0, 0)	(0, 1, 1, 0, 0)	(0, 1, 3, 0, 0)
D	(0, 0, 2, 1, 0)	(0, 0, 2, 0, 0)	(0, 0, 1, 0, 0)	(0, 0, 3, 0, 0)
L	(0, 0, 2, 1, 0)	(0, 0, 2, 0, 0)	(0, 0, 1, 0, 0)	(0, 0, 3, 0, 0)
R	(1, 0, 2, 1, 0)	(1, 0, 2, 0, 0)	(1, 0, 2, 0, 1)	(1, 0, 3, 0, 0)